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(54) PHARMACEUTICAL PREPARATION COMPRISING CEPHALOSPORIN HAVING CATECHOL GROUPS

PHARMAZEUTISCHE ZUBEREITUNG MIT CEPHALOSPORIN MIT KATECHOLGRUPPEN

PRÉPARATION PHARMACEUTIQUE COMPRENANT UNE CÉPHALOSPORINE AYANT DES GROUPES CATÉCHOL

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(56) References cited:
EP-A1- 2 341 053 **WO-A1-2010/050468**
JP-A- H0 459 730 **JP-A- S6 045 514**
JP-A- S6 317 827 **JP-A- H01 216 996**
JP-A- H03 173 822 **JP-A- H03 264 531**
JP-A- H03 264 531 **JP-A- H06 122 630**

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Description

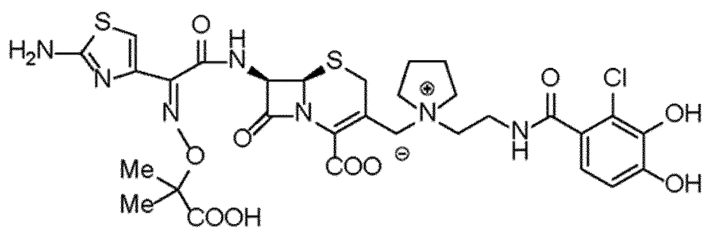
Technical Field

[0001] The pharmaceutical preparation of the present invention relates to a stable pharmaceutical preparation comprising the cephem compound, which has strong antibacterial activity, especially, to gram negative bacteria producing β-lactamase.

Background Art

[0002] Various β-lactam medicines have been developed until now, and β-lactam medicines are very important antibacterial medicines clinically. However, the bacterial strains which have acquired tolerance to β-lactam medicines by producing β-lactamase are increased. The compound represented by the following formula (I), its pharmaceutically acceptable salt or solvate thereof was discovered as an agent to exhibiting strong antibacterial spectrum against several bacteria including gram negative bacteria and/or gram positive bacteria, and the above problem was solved (Patent Document 1).
Formula (I):

[Chemical formula 1]



(I)

[0003] However, when temporal stability test was performed on the compound represented by formula (I), its pharmaceutically acceptable salt or solvate thereof , it was found that the analogue of the compound is produced and the residual ratio of the compound (I) decreased.

[0004] As examples of the pharmaceutical preparation which stabilizes an antibiotic such as a cephem compound, the pharmaceutical preparation containing lactose and sodium chloride as an additive (Patent Document 2), the pharmaceutical preparation containing lactose, sodium chloride and citrate (Patent Document 3), the pharmaceutical preparation containing glucose (Patent Documents 4 and 5), the pharmaceutical preparation containing one or more selected from glucose, fructose and maltose, and alkali metal salt (Patent Document 6), the pharmaceutical preparation containing halogenide (Patent Documents 7), the pharmaceutical preparation containing maltose and sodium chloride (Patent Documents 8), the pharmaceutical preparation containing the salt, the cation of the salt is one or more selected from one or more selected from the group consisting of sodium, calcium and magnesium, the anion of the salt is one or more selected from the group consisting of chloride, fluoride and iodide (Patent Document 9) and the pharmaceutical preparation containing sodium chloride and sucrose (Non-Patent Document 1) are disclosed. However, the pharmaceutical preparation capable of satisfying stability for the compound represented by above formula (I), its pharmaceutically acceptable salt or solvate thereof cannot be prepared by these documents.

[0005] As examples of the stable injectable preparation, the pharmaceutical preparation containing alatrofloxacin as an antibiotic, sodium chloride and sucrose, (Patent Document 10), and the pharmaceutical preparation containing nicardipine hydrochloride as an active ingredient, sodium gluconate, and sodium chloride (Patent Document 11) are disclosed. However, there are great differences on the chemical structure among alatrofloxacin, nicardipine hydrochloride and the compound represented by formula (I), its pharmaceutically acceptable salt or solvate thereof, and these documents cannot contribute the stability of the compound represented by above formula (I), its pharmaceutically acceptable salt or solvate thereof.

Prior Art

Patent Document

[0006]

Patent Document 1: International publication pamphlet WO2010/050468
Patent Document 2: JP-A No.63-17827
Patent Document 3: JP A No. 3-173822
Patent Document 4: JP A No.58-113125
Patent Document 5: JP A No. 7-82149
Patent Document 6: JP A No.60-45514
Patent Document 7: JP A No.2004-269401
Patent Document 8: JP A No.2009-286738
Patent Document 9: JP A No.63-208522
Patent Documents 10: JP A No.2000-219628
Patent Documents 11: JP A No.2001-316266

[0007] Non-Patent Document 1: Yakugakuzasshi Vol. 110, No.3 Page 191-201 (1990)
[0008] JP H03 264531 relates to a preparation of (6R,7R) -7- [2 - (2-amino-4-thiazolyl) -2- [Z-[(S)-carboxy - (3,4-dihydroxyphenyl)methyl] oxyimino] acetamido] -3- [2-carboxy -5-methyl -S-triazolo-[1,5 - a] [pyrimidin -7- yl)thiomethyl] -8- oxo-5-thia -1- azabicyclo [4,2,0] oct-2-ene-2-carboxylic acid alkali metal salt of formula (M is H or alkali metal; at least one of M is alkali metal) dissolved together with an alkali metal salt and/or sugar in an aqueous solvent and the solution is freeze- dried. The amount of the alkali metal salt or the sugar is preferably 5-30wt.% or 10-40wt.%, respectively, based on the compound of formula in isolated state. When the alkali metal salt is used in combination with the sugar, the stability of the preparation can be further improved compared with the separate use of the component.
[0009] EP 2 341 053 discloses compositions comprising lyophilized cephalosporin compounds (known as Cefiderocol or S-649266).

Disclosure of Invention

Problems to be solved by the Invention

[0010] Therefore, a stable preparation comprising the compound represented by the above formula (I), its pharmaceutically acceptable salt or solvate thereof has been desired. Means to solve the Problems
[0011] The present inventors intensively studied, and found out that the pharmaceutical preparation having improved temporal stability can be prepared, that is, the pharmaceutical preparation which comprising 1) the compound represented by formula (I), its pharmaceutically acceptable salt or solvate thereof, preferably, its sodium salt represented by formula (II), 2) in a composition as defined by the claims

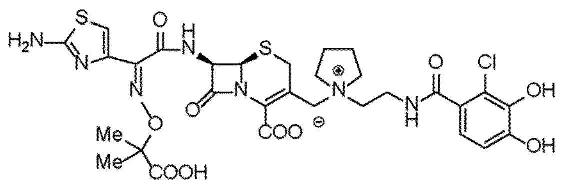
(hereafter, "The pharmaceutical preparation of the present invention")

[0012] That is, the present invention includes:

(1) A pharmaceutical composition characterized by comprising at least the following components:

1) a compound of formula (I):

[Chemical formula 2]

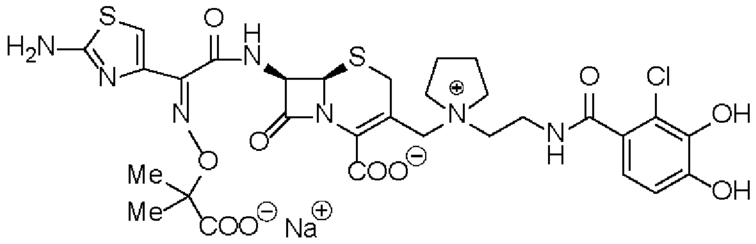


(I)

, its pharmaceutically acceptable salt or solvate thereof,
2) sodium chloride of 1.5 to 4.0 mole equivalent to component 1)
3) sucrose of 1.0 to 3.0 mole equivalent to component 1), and
4) sodium p-toluenesulfonic acid of 0.25 to 2.5 mole equivalent to component 1) and sodium sulphate of 0.05 to 2.0 mole equivalent to component 1).

(2) the pharmaceutical composition according to the above (1), wherein said component 1) is sodium salt of the compound represented by formula (II):

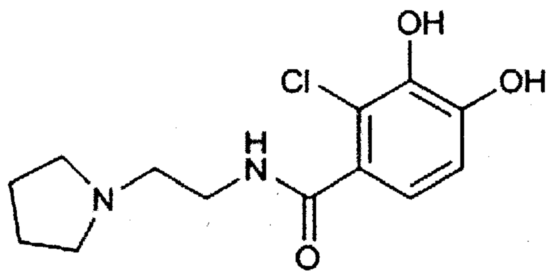
[Chemical formula 3]



(II)

(3) the pharmaceutical composition according to the above (1) or (2), wherein said component 1) is amorphous,
(4) the pharmaceutical composition according to the above (1), wherein the increased amount of the compound represented by formula (III):

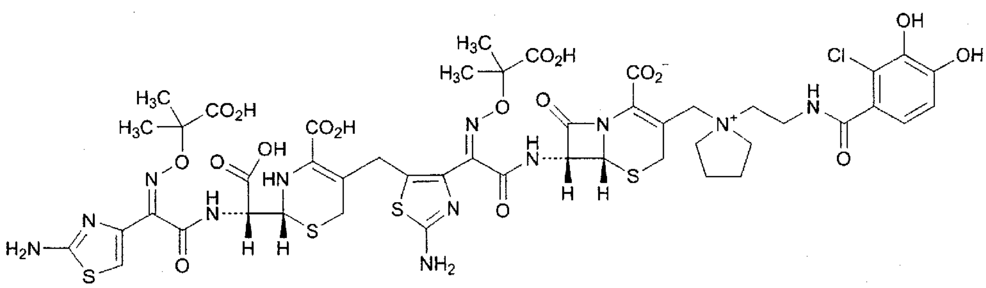
[Chemical formula 4]



(III)

in said pharmaceutical composition is less than 0.4 % from the start of storage under 40°C for two weeks storage,
(5) the pharmaceutical composition according to (1), wherein the increased amount of the compound represented by formula (III) in said pharmaceutical composition is less than 0.05 % from the start of storage under 25°C for two weeks storage,
(6) the pharmaceutical composition according to (1), wherein the increased amount of the compound represented by formula (IV):

[Chemical formula 5]



(IV)

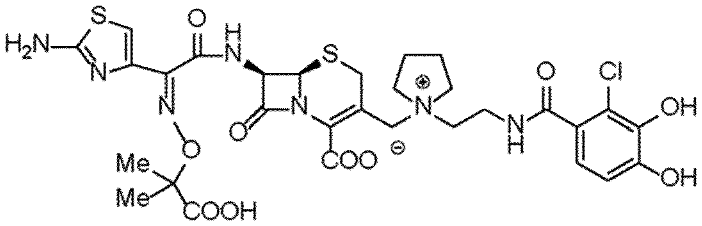
in said pharmaceutical composition is less than 0.05% from the start of storage under 25°C for two weeks storage,
(7) the pharmaceutical composition according to (1), wherein the increased amount of the compound represented by formula (III) in said pharmaceutical composition is less than 0.05 %, and the increased amount of the compound represented by formula (IV) in said pharmaceutical composition is less than 0.05 %,
(8) the pharmaceutical composition according to (1), which further comprises sodium gluconate,
(9) the pharmaceutical composition according to (1), which is a lyophilized product,
(10) the pharmaceutical composition according to (1), which is an injection.

[0013] The present disclosure also provides

(31) a manufacturing method of the pharmaceutical preparation which comprises at least the following steps (the components 2) and 3) are the same meanings in the above (1).):

a) a step of adjusting the liquid comprising the compound represented by formula (I), its pharmaceutically acceptable salt or solvate thereof:

[Chemical formula 8]



(I)

to pH 5 to 6 by alkali materials,
b) a step of mixing the liquid which is prepared by said step a), said component 2) and said component 3), and
c) a step of lyophilizing the mixture prepared by said step b),

(32) the manufacturing method of the pharmaceutical preparation which comprise at least the following steps:

a) the step of adjusting the suspension comprising the compound represented by formula (I), its pharmaceutically acceptable salt or solvate thereof to pH 5 to 6 by sodium hydroxide,
b) the step of mixing the liquid which is prepared by said step a), said component 2) and said component 3), and
c) the step of lyophilizing the mixture prepared by said step b),

(33) The pharmaceutical composition prepared by the manufacturing method of the pharmaceutical preparation according to the above (31) or (32).

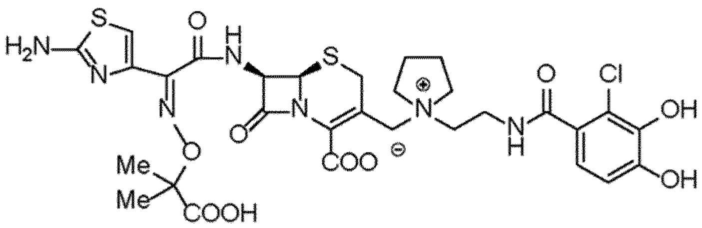
Effect of the Invention

[0014] When the temporal stability test of the pharmaceutical preparation of the present invention was conducted, the stability of the compound represented by the following formula (I), its pharmaceutically acceptable salt or solvate thereof was found to be improved.

Best Mode for Carrying out the Invention

[0015] In the specification, the active ingredient in the pharmaceutical preparation of the present invention is represented by formula (I):

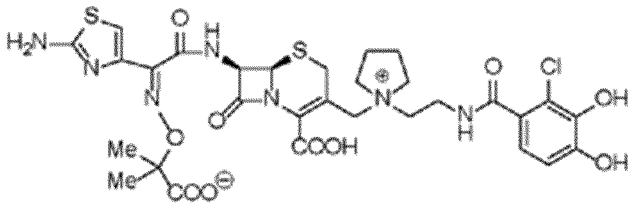
[Chemical formula 9]



(I)

But the compound substantially can also become the state on the compound of formula (I'),

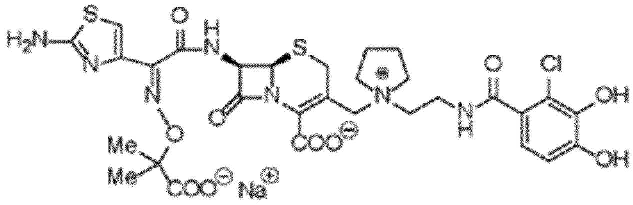
[Chemical formula 10]



(I')

so the compounds of the both chemical structures are included in the present invention. The sodium salt of the compound represented by formula (I) includes

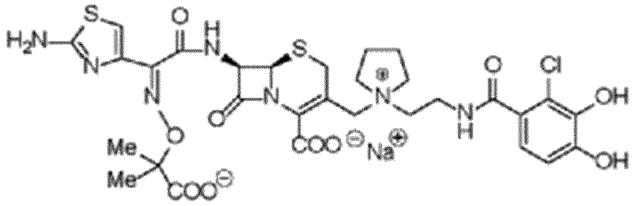
[Chemical formula 11]



(II)

and

[Chemical formula 12]



(II')

[0016] The compound represented by formula (I), its pharmaceutically acceptable salt or solvate thereof, and the sodium salt represented by formula (II) may be amorphous (non-crystalline). Moreover, the molecular weight of the compound represented by formula (I) is 752.21 and that of the sodium salt of the compound represented by formula (II) is 774.20.

[0017] The pharmaceutical preparation according to the invention is defined by the claims, while a further general disclosure comprises one or more selected from the group consisting of alkali metal chloride, alkali earth metal chloride, transition metal chloride and magnesium chloride, and sugar and /or sugar alcohol to stabilize the compound represented by formula (I), its pharmaceutically acceptable salt or solvate thereof and the sodium salt of the compound represented by formula (II).

[0018] As alkali metal chloride, those stabilize the compound represented by formula (I), its pharmaceutically acceptable salt or solvate thereof and the sodium salt of the compound represented by formula (II), and described in Japanese Pharmacopoeia, Pharmaceutical Standards outside the Japanese Pharmacopoeia, Japanese Pharmaceutical Excipients and Japanese Standard of Food Additives may be used. Examples of alkali metal chloride include sodium chloride, lithium chloride, potassium chloride, preferably sodium chloride.

[0019] As alkali earth metal chloride, those stabilize the compound represented by formula (I), its pharmaceutically acceptable salt or solvate thereof and the sodium salt of the compound represented by formula (II), and described in Japanese Pharmacopoeia, Pharmaceutical Standards outside the Japanese Pharmacopoeia, Japanese Pharmaceutical Excipients and Japanese Standard of Food Additives may be used. Examples of alkali earth metal chloride include barium chloride, calcium chloride, preferably calcium chloride.

[0020] As transition metal chloride, those stabilize the compound represented by formula (I), its pharmaceutically acceptable salt or solvate thereof and the sodium salt of the compound represented by formula (II), and described in Japanese Pharmacopoeia, Pharmaceutical Standards outside the Japanese Pharmacopoeia, Japanese Pharmaceutical represented by formula (I), its pharmaceutically acceptable salt or solvate thereof and the sodium salt of the compound represented by formula (II) may be used. Examples of these combinations disclosed but not according to the invention include a) the compound represented by formula (I), its pharmaceutically acceptable acid additive salt or a solvate thereof, sodium chloride, sucrose, sodium p-toluene sulfonate and sodium gluconate, b) the compound represented by formula (I), its pharmaceutically acceptable acid additive salt or a solvate thereof, sodium chloride, sucrose, sodium sulfate and sodium gluconate, c) the compound represented by formula (I), its pharmaceutically acceptable acid additive salt or a solvate thereof, sodium chloride, sucrose, sodium p-toluene sulfonate, sodium sulfate and sodium gluconate, d) sodium salt of the compound represented by formula (II), sodium chloride, sucrose, sodium p-toluene sulfonate and sodium gluconate, e) sodium salt of the compound represented by formula (II), sodium chloride, sucrose and sodium sulfate and sodium gluconate, f) sodium salt of the compound represented by formula (II), sodium chloride, sucrose, sodium p-toluene sulfonate, sodium sulfate and sodium gluconate, g) the compound represented by formula (I), its pharmaceutically acceptable acid additive salt or a solvate thereof, magnesium chloride, sucrose, sodium p-toluene sulfonate and sodium gluconate, h) the compound represented by formula (I), its pharmaceutically acceptable acid additive salt or a solvate thereof, magnesium chloride, sucrose, sodium sulfate and sodium gluconate, i) the compound represented by formula (I), its pharmaceutically acceptable acid additive salt or a solvate thereof, magnesium chloride, sucrose, sodium p-toluene sulfonate, sodium sulfate and sodium gluconate, j) sodium salt of the compound represented by formula (II), magnesium chloride, sucrose, sodium p-toluene sulfonate and sodium gluconate, k) sodium salt of the compound represented by formula (II), magnesium chloride, sucrose, sodium sulfate and sodium gluconate, l) sodium salt of the compound represented by formula (II), magnesium chloride, sucrose, sodium p-toluene sulfonate, sodium sulfate and sodium gluconate. Preferably, c),f), and more preferably, f).

[0040] When the pharmaceutical preparation comprises sodium chloride as alkali metal chloride, sucrose as sugar and/or sugar alcohol, sodium p-toluene sulfonate as alkali metal salt of p-toluene sulfonate and sodium sulfate as alkali metal salt of sulfuric acid to the compound represented by formula (I), its pharmaceutically acceptable salt or solvate thereof, a content of sodium chloride is 1.5 to 4.0 molar equivalent, sucrose is 1.0 to 3.0 molar equivalent, sodium p-toluene sulfonate is 0.75 to 2.0 molar equivalent, sodium sulfate is 0.1 to 1.0 molar equivalent and sodium gluconate is 0.1 to 0.5 molar equivalent.

[0041] When the pharmaceutical preparation comprises sodium chloride as alkali metal chloride, sucrose as sugar and/or sugar alcohol, sodium p-toluene sulfonate and sodium sulfate to the sodium salt of the compound represented by formula (II), a content of sodium chloride is 1.5 to 4.0 molar equivalent, sucrose is 1.0 to 3.0 molar equivalent, sodium p-toluene sulfonate is 0.75 to 2.0 molar equivalent, sodium sulfate is 0.1 to 1.0 molar equivalent and sodium gluconate is 0.1 to 0.5 molar equivalent.

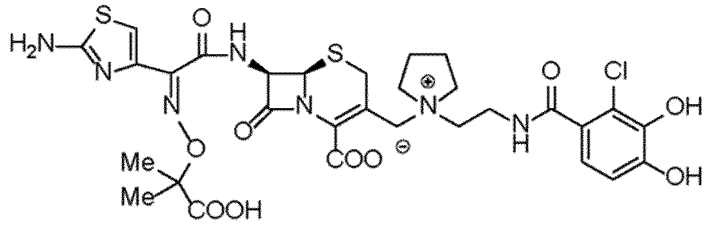
[0042] The pharmaceutical preparation of the present invention, those the compound represented by formula (I), its pharmaceutically acceptable salt or solvate thereof and the sodium salt of the compound represented by formula (II) and additives is dissolved or suspended in water and dried. The drying method, those stabilize the compound represented by formula (I), its pharmaceutically acceptable salt or solvate thereof and the sodium salt of the compound represented by formula (II) may be used. Examples of drying method involve vacuum drying method by evaporator, spray-drying method and lyophilized method, preferably lyophilized method, the desirable preparation of the present invention is lyophilized product.

[0043] The pharmaceutical preparation of the present invention comprising the compound represented by formula (I), its pharmaceutically acceptable salt or solvate thereof and the sodium salt of the compound represented by formula (II) can be administered as the injection.

[0044] Also disclosed but not claimed is a manufacturing process on the pharmaceutical preparation of the present disclosure comprising at least the following steps:

a) a step of adjusting the liquid comprising the compound represented by formula (I), its pharmaceutically acceptable salt or a solvate thereof:

[Chemical formula 13]



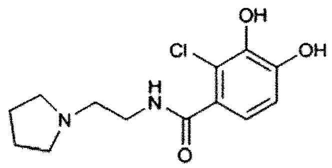
(I)

to pH 5 to 6 by alkali materials,

as injectable distilled water, physiological saline or glucose solution and dissolving the preparation, timely. The pharmaceutical preparation of the present disclosure after lyophilizing is administered after adding a solution such as a distilled water for injection, normal saline solution or glucose solution at the time of use to dissolve. The pharmaceutical composition of the present invention exhibits a strong antibacterial spectrum against Gram-positive bacteria and Gram-negative bacteria, especially β-lactamase producing Gram-negative bacteria, and it does not exhibit cross-resistance with existing cephem drugs and carbapenems.

[0053] The analog is sometimes produced in the composition comprising the compound represented by the above formula (I). When the composition is stored temporally, the amount of the analog increases more. As the analog, mainly,

[Chemical formula 14]

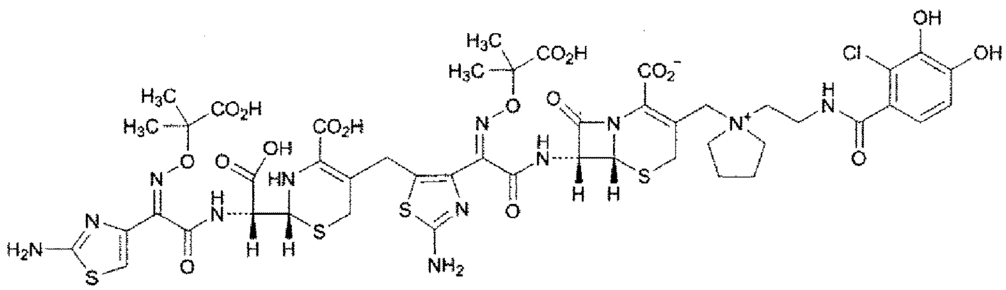


(III)

the compound represented by formula (III) arises. It is preferable that the increased amount of the compound represented by formula (III) is low under the consideration of toxicity. When the composition of the present invention is preserved at 40°C for two weeks, the increased amount from the start of test of the compound represented by formula (III) in said medicinal composition is less than 0.45 %, preferably, less than 0.425%, more preferably, less than 0.4%. Moreover, when the composition of the present invention was preserved at 25°C for two weeks, the increased amount from the start of test of the compound represented by formula (III) in said medicinal composition is less than 0.06 %, preferably, less than 0.055%, more preferably, less than 0.05%, especially preferably, less than 0.04%.

[0054] The dimer of the compound represented by formula (I) is produced in the composition comprising the compound represented by formula (I), and when the composition is preserved temporally, the amount of said dimer increases. As the dimer, mainly, the compound represented by formula (IV):

[Chemical formula 15]



(IV)

arises. It is preferable that the increased amount of compound represented by formula (IV) is low under the consideration of toxicity. When the composition of the present invention was preserved at 25 °C for two weeks, the increased amount from the start of test of the compound represented by formula (IV) in said medicinal composition is less than 0.06 %, preferably, less than 0.055%, more preferably, less than 0.05%, especially preferably, less than 0.04%.

Examples

[0055] The present invention will be explained in more detail below by way of Examples, and Comparative Examples, but these do not limit the present invention.

1. The manufacturing method of the compound represented by formula (I), its pharmaceutically acceptable salt or solvate thereof (1.3 tosilate salt/0.4 sulfuric acid salt)

<Preparation of a seed crystal A of 2 mole equivalents of p-toluenesulfonic acid salt of the compound (IA)>

[0056] The compound (I) (100mg) was dissolved in 1.0 mol/L p-toluenesulfonic acid solution (2mL) at room temperature

Availability in the industry

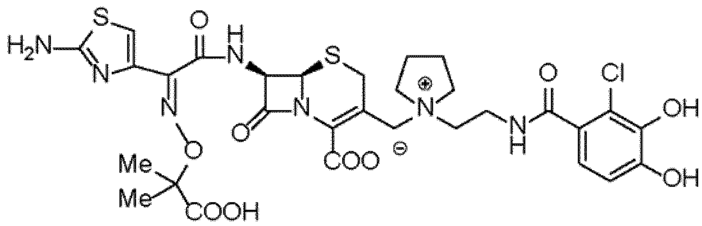
[0119] The pharmaceutical preparation of the present invention is useful as the injection of antibiotics which has strong antibacterial activity and high stability against gram negative bacteria which product β-lactamase.

Claims

1. A pharmaceutical composition **characterized by** comprising at least the following components:

    1) a compound of formula (I):

    [Chemical formula 2]

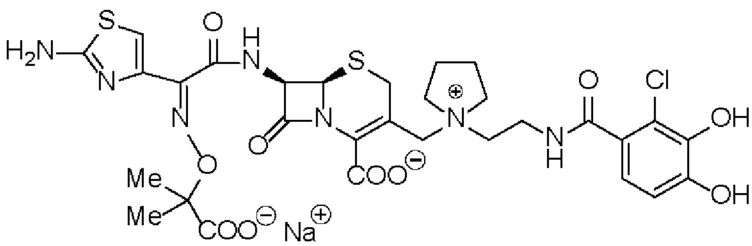
    

    (I)

    , its pharmaceutically acceptable salt or solvate thereof,
    2) sodium chloride of 1.5 to 4.0 mole equivalent to component 1),
    3) sucrose of 1.0 to 3.0 mole equivalent to component 1), and
    4) sodium p-toluenesulfonic acid of 0.25 to 2.5 mole equivalent to component 1) and sodium sulphate of 0.05 to 2.0 mole equivalent to component 1).

2. The pharmaceutical composition according to claim 1, wherein said component 1) is sodium salt of the compound represented by formula (II):

    [Chemical formula 3]

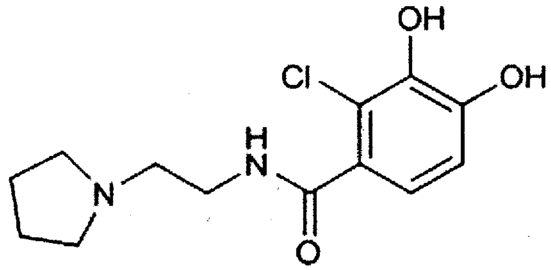
    

    (II)

3. The pharmaceutical composition according to claim 1 or 2, wherein said component 1) is amorphous.

4. The pharmaceutical composition according to claim 1, wherein the increased amount of the compound represented by formula (III):

[Chemical formula 4]



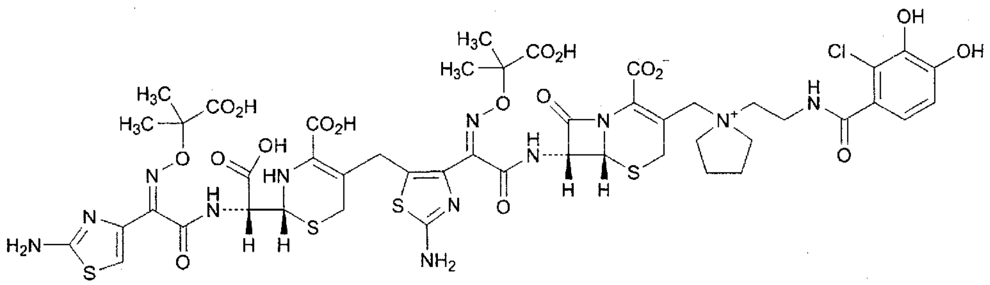
(III)

in said pharmaceutical composition is less than 0.4 % from the start of storage under 40°C for two weeks storage.

5. The pharmaceutical composition according to claim 1, wherein the increased amount of the compound represented by formula (III) in said pharmaceutical composition is less than 0.05 % from the start of storage under 25°C for two weeks storage.

6. The pharmaceutical composition according to claim 1, wherein the increased amount of the compound represented by formula (IV):

[Chemical formula 5]



(IV)

in said pharmaceutical composition is less than 0.05% from the start of storage under 25°C for two weeks storage.

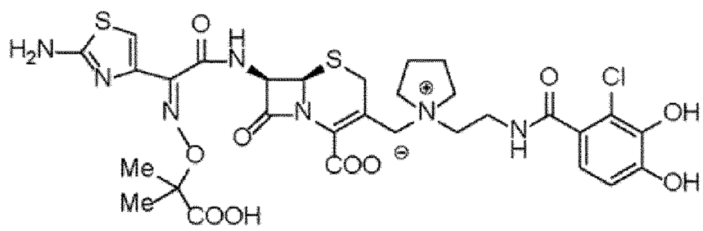
7. The pharmaceutical composition according to claim 1, wherein the increased amount of the compound represented by formula (III) in said pharmaceutical composition is less than 0.05 %, and the increased amount of the compound represented by formula (IV) in said pharmaceutical composition is less than 0.05 % from the start of storage under 25°C for two weeks storage.

Patentansprüche

1. Pharmazeutische Zusammensetzung, **dadurch gekennzeichnet, dass** sie mindestens die folgenden Komponenten umfasst:

1) eine Verbindung der Formel (I):

[Chemische Formel 2]

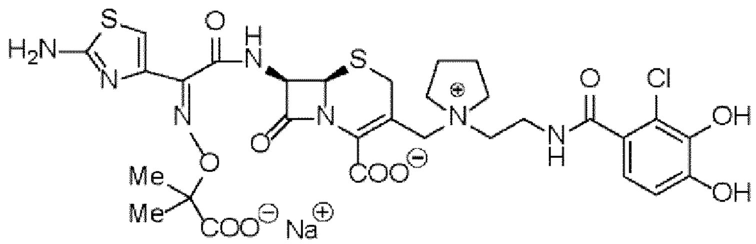


(I)

,

ihr pharmazeutisch akzeptables Salz oder Solvat davon,
2) Natriumchlorid in einer Menge von 1,5 bis 4,0 Moläquivalenten der Komponente 1),
3) Saccharose in einer Menge von 1,0 bis 3,0 Moläquivalenten der Komponente 1), und
4) Natrium-p-Toluolsulfonsäure in einer Menge von 0,25 bis 2,5 Moläquivalenten der Komponente 1) und Natriumsulfat in einer Menge von 0,05 bis 2,0 Moläquivalenten der Komponente 1).

2. Pharmazeutische Zusammensetzung nach Anspruch 1, bei der die Komponente 1) Natriumsalz der Verbindung, dargestellt durch die Formel (II), ist:

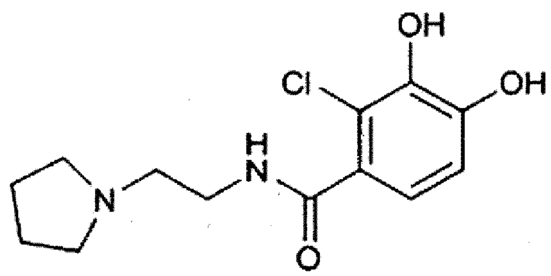


(II)

3. Pharmazeutische Zusammensetzung nach Anspruch 1 oder 2, bei der die Komponente 1) amorph ist.

4. Pharmazeutische Zusammensetzung nach Anspruch 1, bei der ab dem Beginn der Lagerung bei 40 °C für zwei Wochen Lagerung die Zunahme der Menge der durch die Formel (III) dargestellten Verbindung:

[Chemische Formel 4]



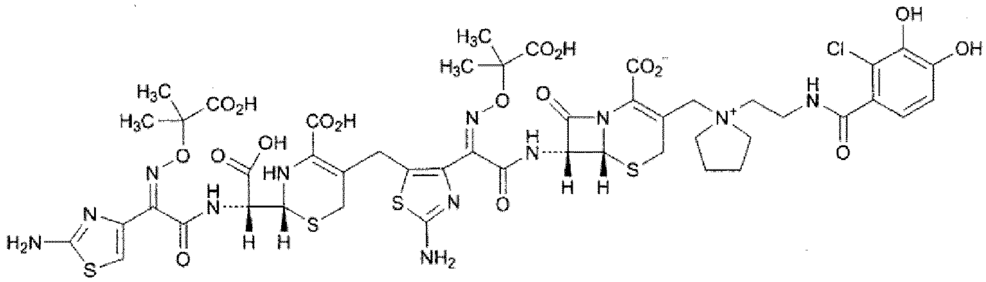
(III)

in der pharmazeutischen Zusammensetzung weniger als 0,4 % beträgt.

5. Pharmazeutische Zusammensetzung nach Anspruch 1, bei der ab dem Beginn der Lagerung bei 25°C für zwei Wochen Lagerung die Zunahme der durch die Formel (III) dargestellten Verbindung in der pharmazeutischen Zusammensetzung weniger als 0,05 % beträgt.

6. Pharmazeutische Zusammensetzung nach Anspruch 1, bei der ab Beginn der Lagerung bei 25°C für zwei Wochen Lagerung die Zunahme der Menge der Verbindung der Formel (IV):

[Chemische Formel 5]



(IV)

in der pharmazeutischen Zusammensetzung weniger als 0,05% beträgt.

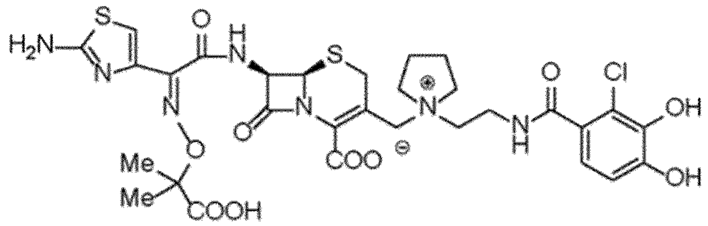
7. Pharmazeutische Zusammensetzung nach Anspruch 1, bei der ab dem Beginn der Lagerung bei 25°C für zwei Wochen Lagerung die Zunahme der durch Formel (III) dargestellten Verbindung in der pharmazeutischen Zusammensetzung weniger als 0,05 % und die Zunahme der durch die Formel (IV) dargestellten Verbindung in der pharmazeutischen Zusammensetzung weniger als 0,05 % beträgt.

Revendications

1. Composition pharmaceutique **caractérisée en ce qu'**elle comprend au moins les composants suivants :

1) un composé de formule (I) :

[Formule chimique 2],



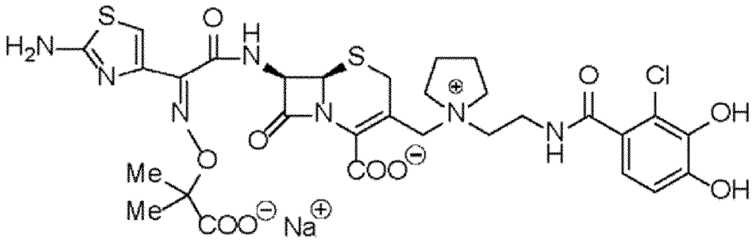
(I)

,

son sel pharmaceutiquement acceptable ou son solvate,
2) du chlorure de sodium de 1,5 à 4,0 moles d'équivalent au composant 1),
3) du saccharose de 1,0 à 3,0 moles d'équivalent au composant 1), et
4) de l'acide p-toluènesulfonique de sodium de 0,25 à 2,5 moles d'équivalent au composant 1) et du sulfate de sodium de 0,05 à 2,0 moles d'équivalent au composant 1).

2. Composition pharmaceutique selon la revendication 1,
dans laquelle ledit composant 1) est un sel de sodium du composé représenté par la formule (II) :

[Formule chimique 3]

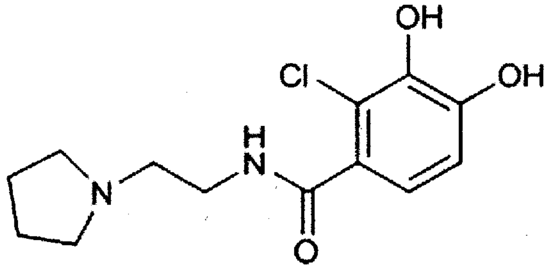


(II)

3. Composition pharmaceutique selon la revendication 1 ou 2, dans laquelle ledit composant 1) est amorphe.

4. Composition pharmaceutique selon la revendication 1, dans laquelle la quantité accrue du composé représenté par la formule (III) :

[Formule chimique 4]



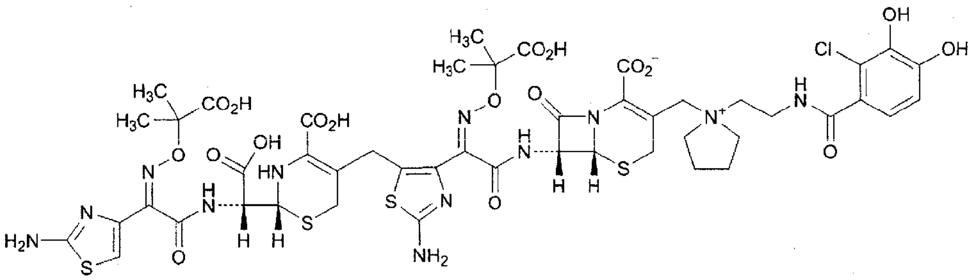
(III)

dans ladite composition pharmaceutique est inférieure à 0,4 % à partir du début du stockage sous 40°C pendant deux semaines de stockage.

5. Composition pharmaceutique selon la revendication 1, dans laquelle la quantité accrue du composé représenté par la formule (III) dans ladite composition pharmaceutique est inférieure à 0,05 % depuis le début du stockage sous 25°C pendant deux semaines de stockage.

6. Composition pharmaceutique selon la revendication 1, dans laquelle la quantité accrue du composé représenté par la formule (IV) :

[Formule chimique 5]



(IV)

dans ladite composition pharmaceutique est inférieure à 0,05 % à partir du début du stockage à moins de 25°C pendant deux semaines de stockage.

7. Composition pharmaceutique selon la revendication 1, dans laquelle la quantité accrue du composé représenté par la formule (III) dans ladite composition pharmaceutique est inférieure à 0,05 %, et la quantité accrue du composé représenté par la formule (IV) dans ladite composition pharmaceutique est inférieure à 0,05 % à partir du début du stockage sous 25°C pendant deux semaines de stockage.